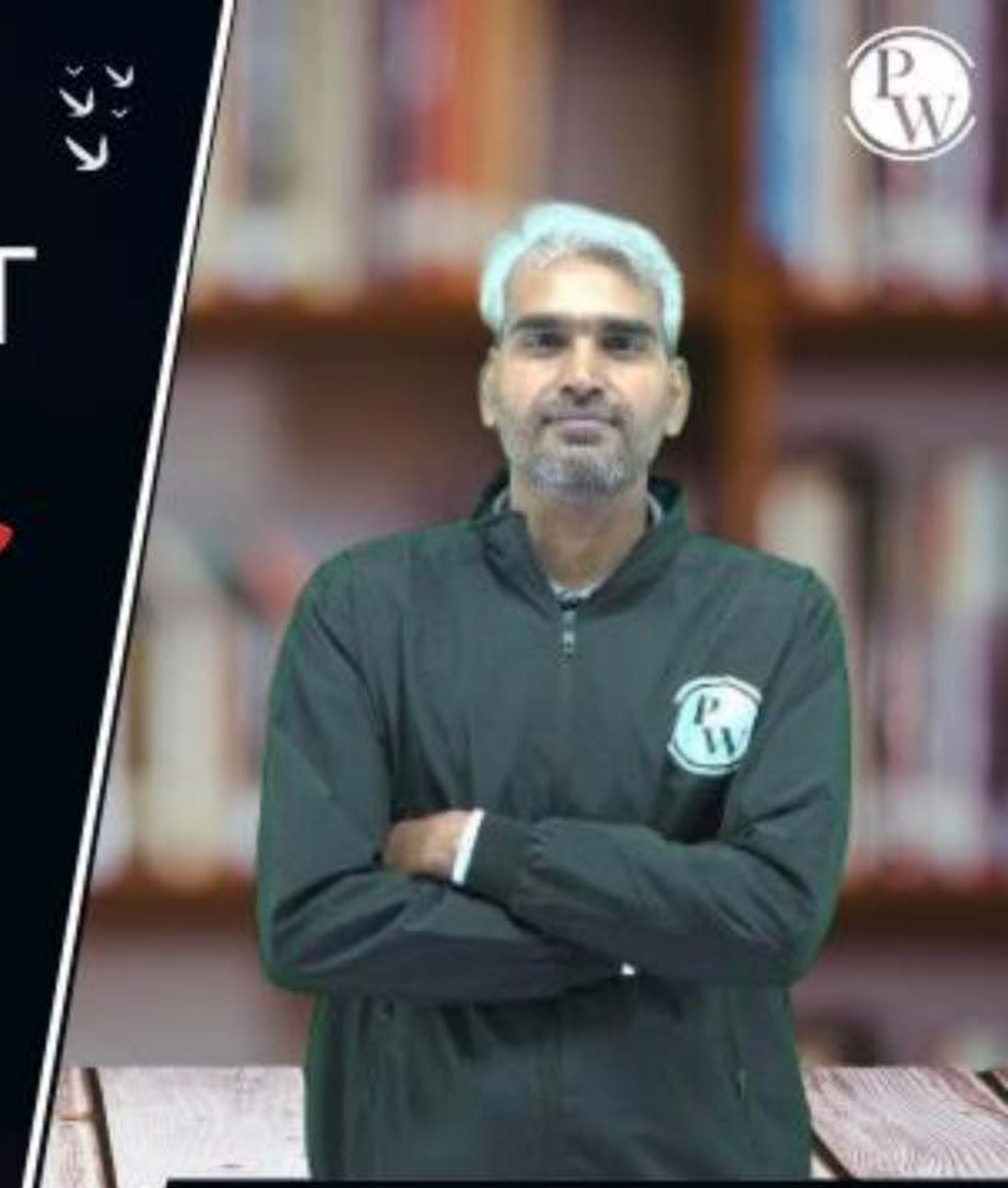
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 17



By- Vishal Sir

Recap of Previous Lecture



Topic — Bounded lattice

Topic — Complement of an element in a lattice

Topic — Complemented lattice

Topic — Sub lattice

Slide

Topics to be Covered



Topic Distributive lattice

Topic Boolean algebra

Slide

Bounded Lattice

→ A lattice in which both minimum as well as maximum element exist is called a bounded lattice



Topic : Bounded Lattice

✓ Let $[L, \vee, \wedge]$ be a lattice

① If there exists an element $I \in L$ s.t. $a \vee I = I \quad \forall a \in L$.
then 'I' is called Universal upper bound
(or) Upper bound of lattice L.

* "I" is actually the maximum element of lattice.

② If there exist an element $O \in L$.
such that, $a \wedge O = O \quad \forall a \in L$
then, 'O' is called Universal lower bound
(or) lower bound of lattice L

* 'O' is actually the minimum element

* A lattice $[L, \vee, \wedge]$ is called a bounded lattice if and only if both 'I' and 'O' exist in the lattice
(or)
* A lattice is called a bounded lattice if and only if both maximum & minimum element exist.

Slide

Note :- ① Every finite lattice is a bounded lattice

② If lattice is an infinite lattice then it may or may not be a bounded lattice

eg: $(Z, \leq)$ and $(N, \leq)$ are example of unbounded lattice where both Z & N are infinite sets

Bounded lattice w.r.t. infinite set

eg: let $A = \{x \mid x \in R \text{ and } 0 \leq x \leq 1\}$

↳ A is an infinite set.

We know $(A, \leq)$ is a lattice where 'A' is an infinite set

but 0 = Minimum element

& 1 = Maximum element

→ Both minimum & maximum element exist so it is a bounded lattice.



Topic : Complement of an element in a lattice

Complement of an element of lattice $[L, \vee, \wedge]$ exists only if lattice is a bounded lattice

P only if Q
|||
if P then Q
|||
$P \longrightarrow Q$

i.e., ① If Complement exists for any element of lattice $[L, \vee, \wedge]$, then it is a bounded lattice

&

② If lattice is not a bounded lattice then Complement does not exist for any element of that lattice

Slide



Topic : Complement of an element in a lattice

Let $[L, \vee, \wedge]$ be a bounded lattice,

Where I = Universal upper bound (i.e. Maximum element of lattice L)

& O = Universal lower bound (i.e., Minimum element of lattice L)

For an element $a \in L$

if there exist some elements $b \in L$

such that (i) $a \vee b = I$ (maximum element)

$lub(a, b) = I$

and (ii) $a \wedge b = O$ (minimum element)

$glb(a, b) = O$

then element a & b are called Complement of each other w.r.t lattice $[L, \vee, \wedge]$

Slide

Note:

① In a bounded lattice,
'I' and '0' are always Complement of each other
and they do not have any other Complement.

i.e.: The only Complement of Maximum element(I) is minimum element(0)
and The only Complement of minimum element(0) is maximum element(I)

② In a bounded lattice,
Complement need not exist for every element
of the lattice, and if exists it need not be unique
{i.e. some elements may have more
than one Complement

↳ In a bounded lattice number of Complements of an element
of the lattice may be '0' or '1' or more



Topic : Complemented lattice

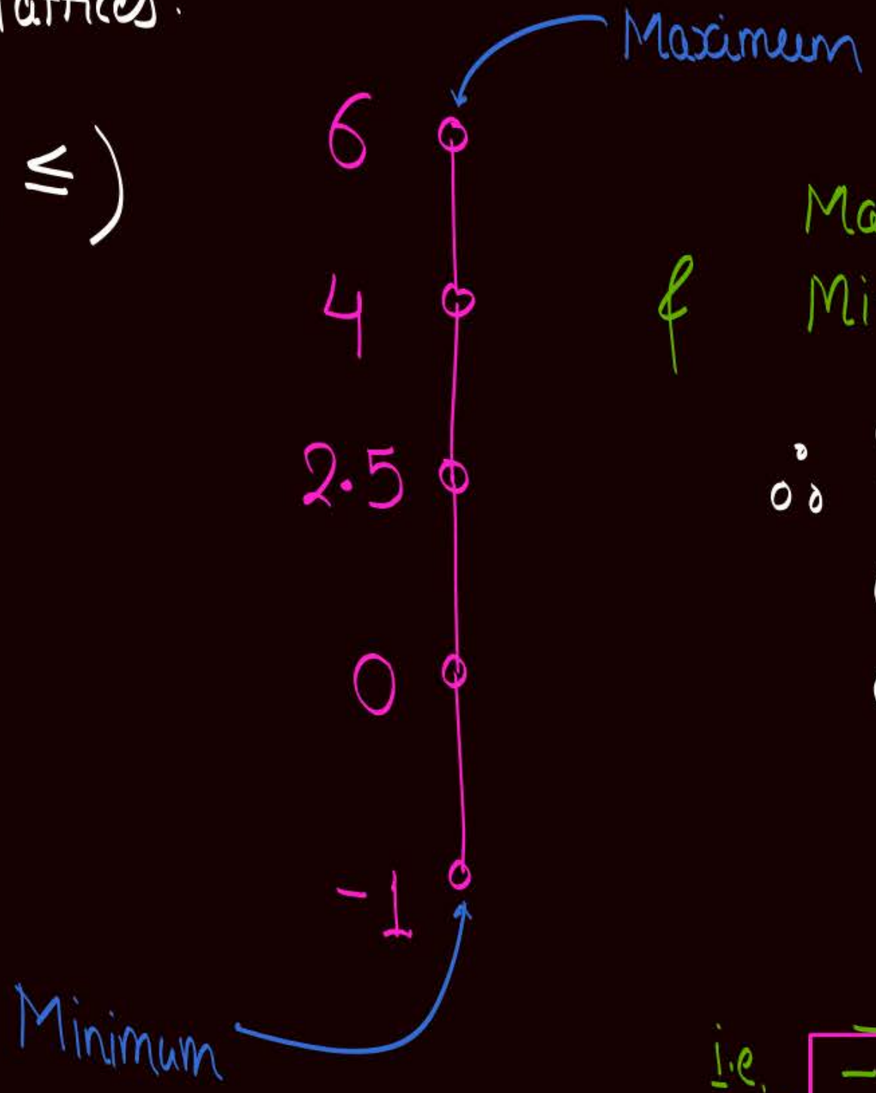
A lattice is called a Complemented lattice if and only if at least one Complement exist for every element of the lattice.

- i.e. in a Complemented lattice every element will have '1' or "more" Complements.

Slide

Find the complement/s of Every element (if Exist) of the following lattices.

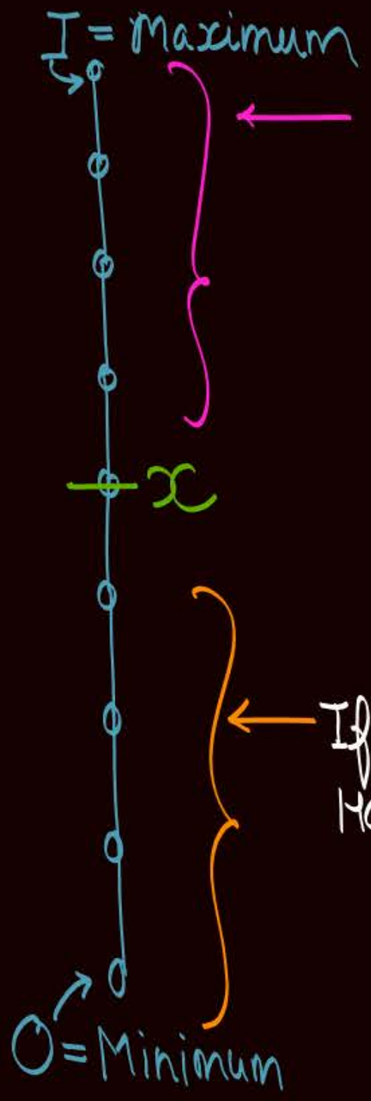
(i) $(\{-1, 0, 2.5, 4, 6\}, \leq)$



Maximum element = 6
&
Minimum element = -1

∴ '-1' & '6' are definitely Complement of each other
&
No other Complement will exist for them

i.e., $\overline{-1} = 6$ & $\overline{6} = -1$



If we choose any element 'a' above 'x' in the Hasse diagram, then $glb(x, a) = x \neq$ Minimum

∴ No such element 'a' can be complement of 'x'

$x \vee x = x \neq$ Maximum

$x \wedge x = x \neq$ Minimum

} ∴ x can not be the complement of 'x' itself.

∴ $\bar{x}$ can not exist

If we choose any element 'b' below 'x' in the Hasse diagram, then $lub(x, b) = x \neq$ Maximum

∴ No such element 'b' can be complement of 'x'

Note:- ① A TOSET is always a lattice, but Every lattice need not be a TOSET.

② If TOSET is a bounded lattice, then

Complement of minimum element = maximum element

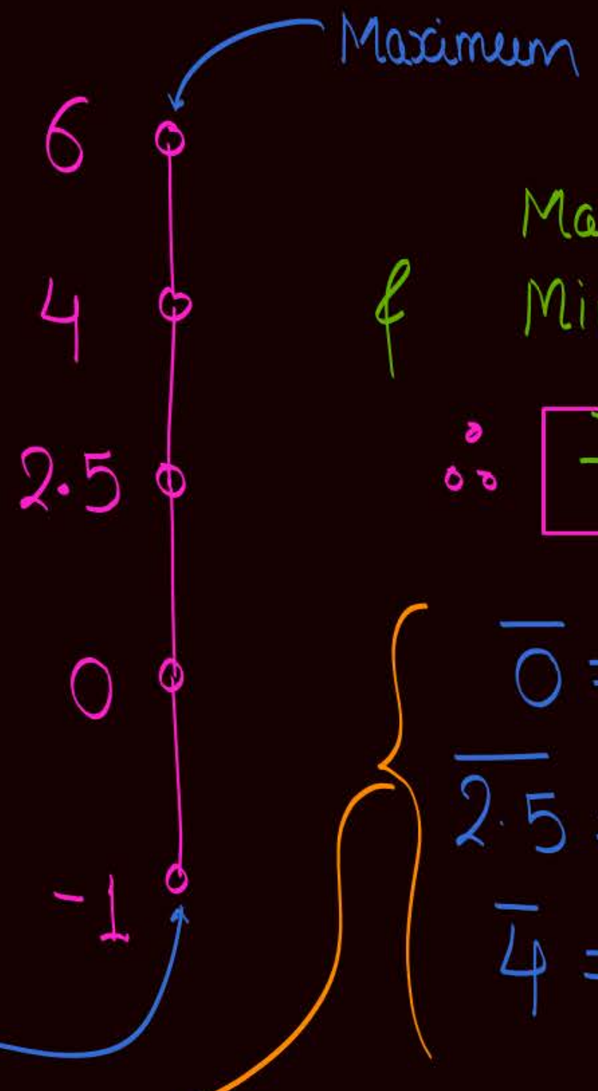
and Complement of maximum element = minimum element

but Complement does not exist for any other element in a lattice w.r.t. TOSET

③ If two elements lie on the same line in a Hasse diagram, then they can not be Complement of each other. Except for minimum & maximum element

Find the complement/s of Every element (if Exist) of the following lattices.

(i) $(\{-1, 0, 2.5, 4, 6\}, \leq)$



Maximum element = 6
&
Minimum element = -1

$\therefore$ $\overline{-1} = 6$ & $\overline{6} = -1$

$\therefore$ Given lattice is not a Complemented lattice.

$\overline{0} = DNE$

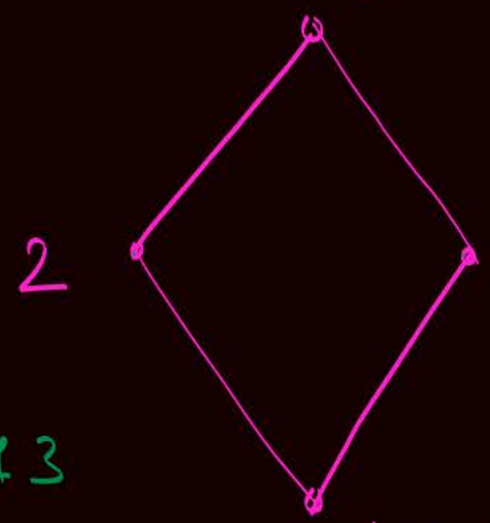
$\overline{2.5} = DNE$

$\overline{4} = DNE$

Find the complement of Every element (if Exist) of the following lattices.

(ii) $(D_6, \div)$

$\{1, 2, 3, 6\}$



$\text{lub}(2,3) = 6 = \text{Maximum}$

&

$\text{glb}(2,3) = 1 = \text{Minimum}$

$\Rightarrow$ 2 & 3 are Complement of each other

$\bar{1} = 6$

$\bar{6} = 1$

$\bar{2} = 3$

$\bar{3} = 2$

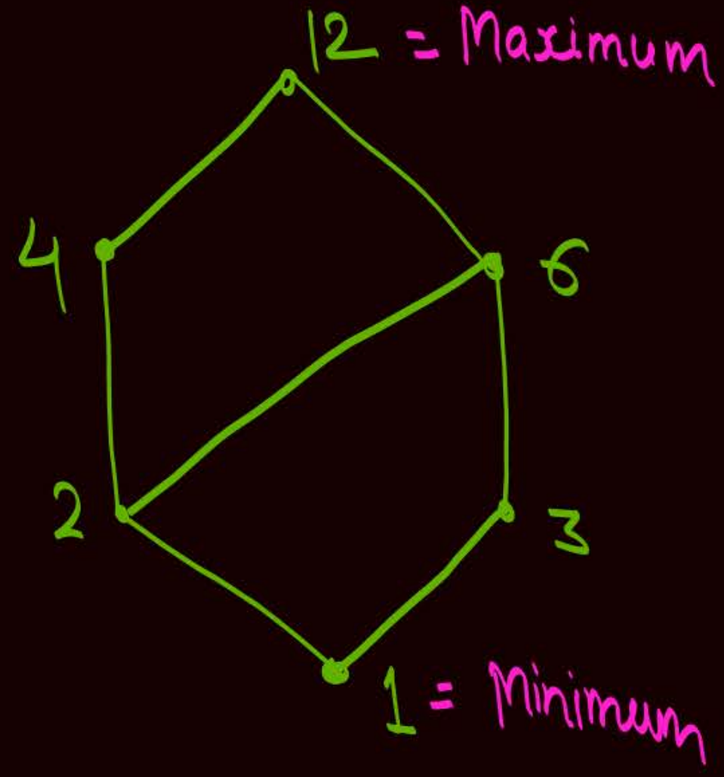
Every element has Exactly '1' complement

$\Downarrow$

$\therefore$ Complemented lattice.

Find the complement of Every element (if Exist) of the following lattices.

(iii) $(D_{12}, \div)$



$\bar{1} = 12$

$\bar{12} = 1$

$\bar{2}$ = Does not exist

$\bar{3} = 4$

$\bar{4} = 3$

$\bar{6}$ = Does not exist

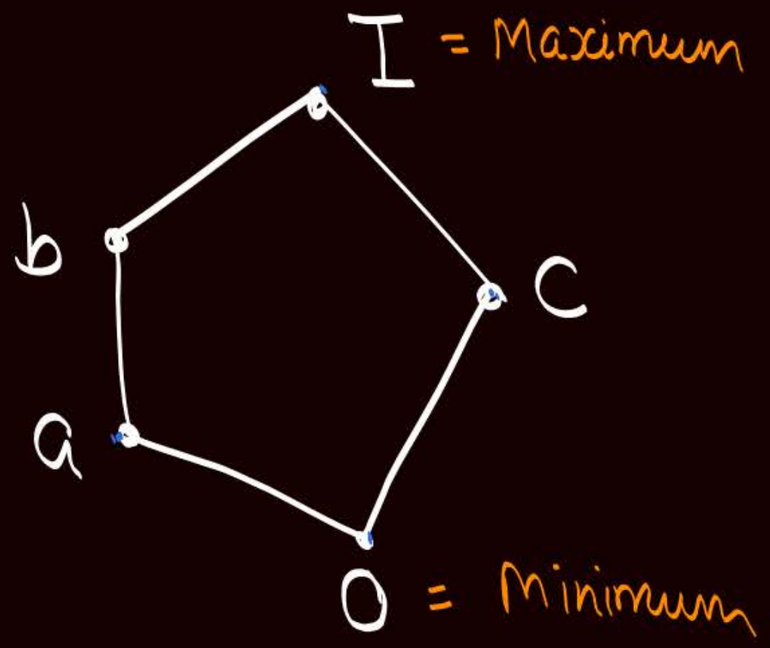
∴ Not a Complemented lattice.

lub(2,3) = 6 ≠ Maximum
∴ 2 & 3 are not Complement

lub(3,4) = 12 = Max
glb(3,4) = 1 = Min
∴ 3 & 4 are Complement of each other

glb(4,6) = 2 ≠ Minimum
∴ 4 & 6 are not Complement

Eg:- Consider the following POSET diagram for a lattice, find the complements for every element.



$\bar{O} = I$

$\bar{I} = O$

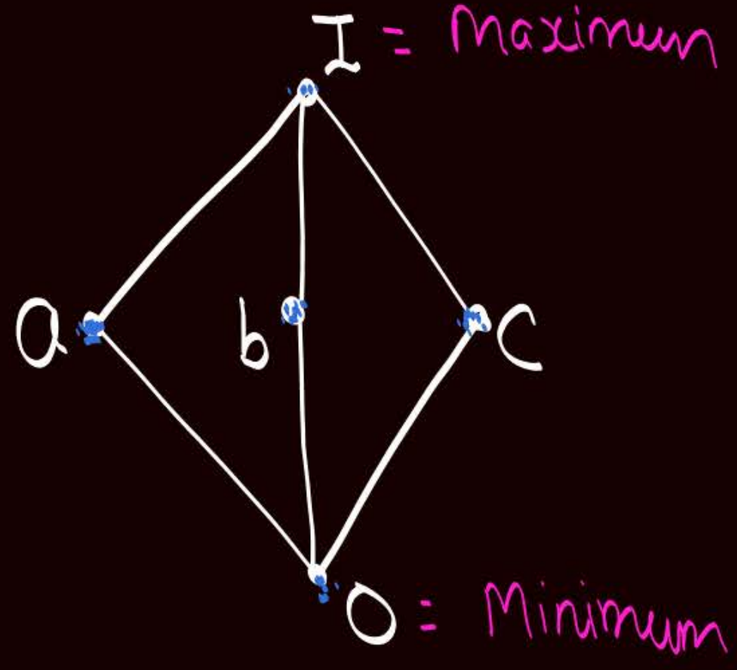
$\bar{a} = c$

$\bar{b} = c$

$\bar{c} = a, b$

∴ it is a Complemented lattice

Eg:- Consider the following POSET diagram for a lattice, find the complements for every element.



$\bar{O} = I$

$\bar{I} = O$

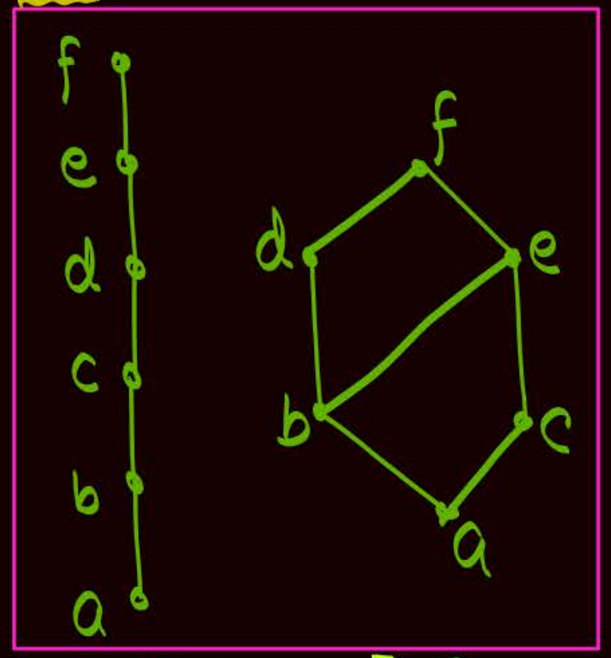
$\bar{a} = b, c$

$\bar{b} = a, c$

$\bar{c} = a, b$

∴ it is a Complemented lattice

Note:-

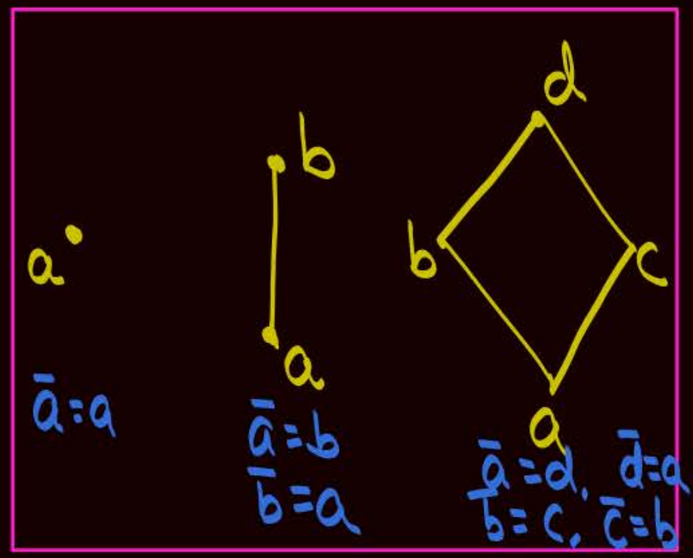


$\bar{a} = f$
$\bar{f} = a$

Complement does not exist for any other element

$\bar{a} = f$
$\bar{f} = a$
$\bar{d} = c$
$\bar{c} = d$
$\bar{b} = DNE$
$\bar{e} = DNE$

These are not Complemented lattice

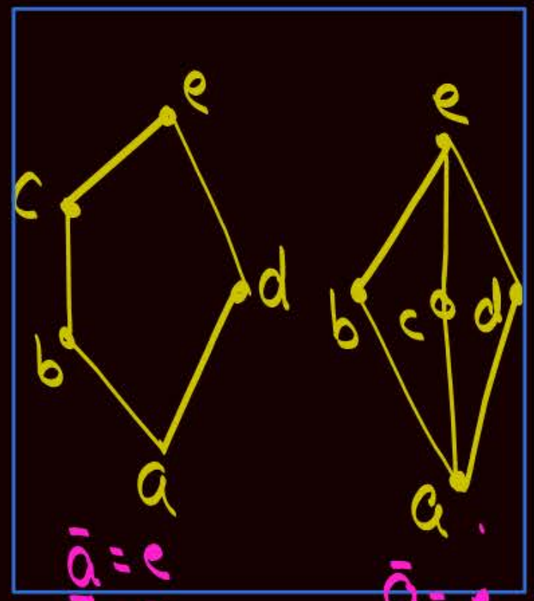


$\bar{a} = a$

$\bar{a} = b$
$\bar{b} = a$

$\bar{a} = d,\ \bar{d} = a$
$\bar{b} = c,\ \bar{c} = b$

Every element has Exactly '1' Complement



$\bar{a} = e$
$\bar{b} = a, d$
$\bar{c} = d$
$\bar{d} = b, c$

$\bar{a} = e$
$\bar{b} = a$
$\bar{c} = c, d$
$\bar{d} = b, d$
$\bar{e} = b, c$

All are Complemented lattice



Topic : Sublattice

Let $[L, \vee, \wedge]$ be a lattice, and M be any subset of set L

'M' is a sub-lattice of lattice 'L' if and only if,

(i) $[M, \vee, \wedge]$ is a lattice

{ i.e. lub as well as glb must exist for Every pair of elements of set M in set M itself }

and (ii) lub and glb for Every pair of elements in lattice $[M, \vee, \wedge]$ must be Exactly same as lub and glb for that pair of elements in lattice $[L, \vee, \wedge]$

i.e. $\forall a, b \in M$

$\text{lub}(a,b)$ in M = $\text{lub}(a,b)$ in L

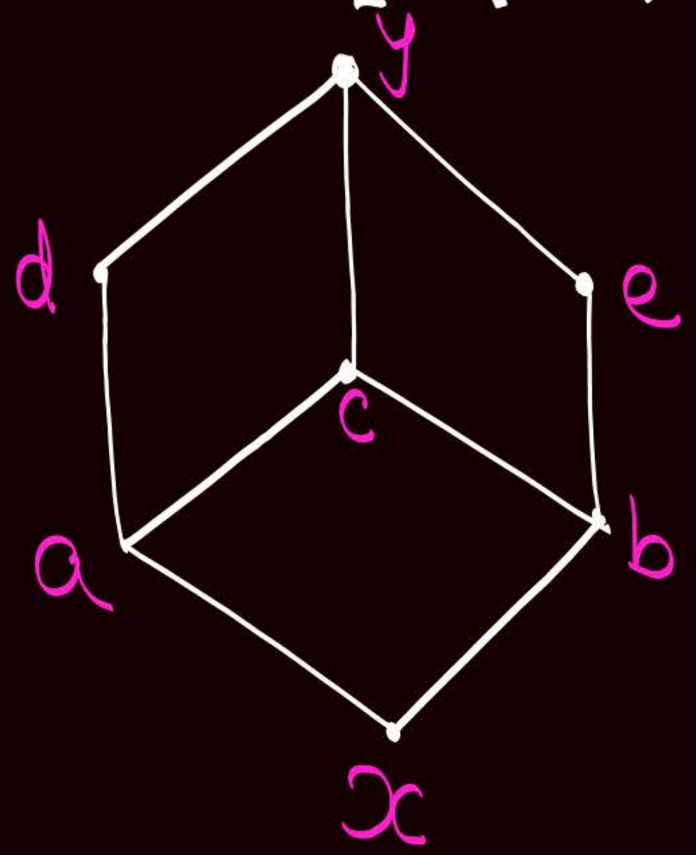
&

$\text{glb}(a,b)$ in M = $\text{glb}(a,b)$ in L

Slide

eg Consider the following Lattice 'L'

"L" = {x, y, a, b, c, d, e}



Which of the following is/are Sublattice of lattice L

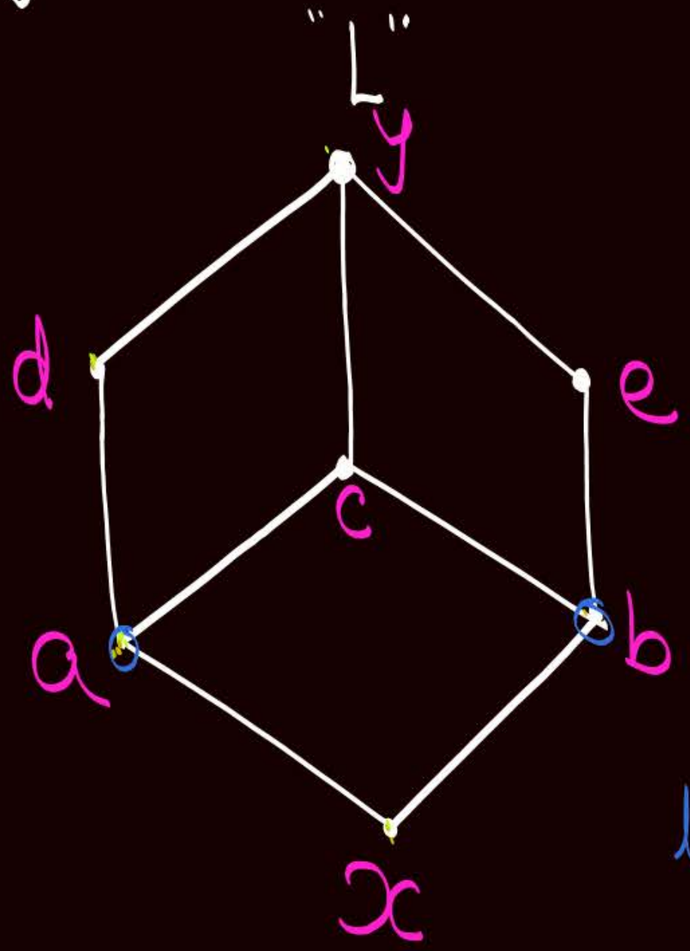
(i) $L_1 = \{x, a, b, y\}$

(ii) $L_2 = \{x, c, d, y\}$

(iii) $L_3 = \{x, a, c, y\}$

(iv) $L_4 = \{x, a, e, y\}$

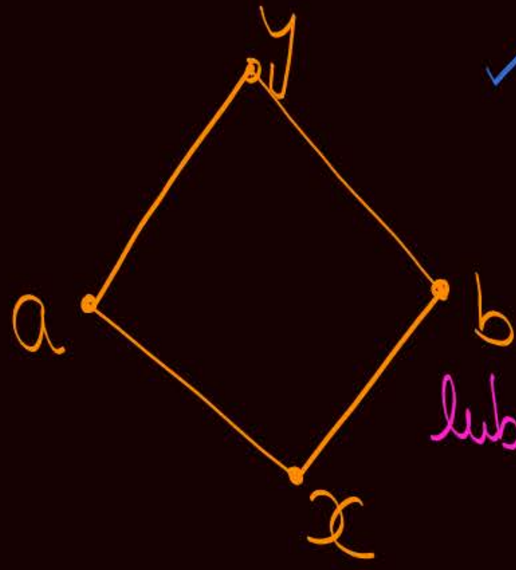
eg Consider the following lattice 'L'



lub(a,b) = c

Which of the following is/are sublattices of lattice L

(i) $L_1 = \{x, a, b, y\}$



L_1

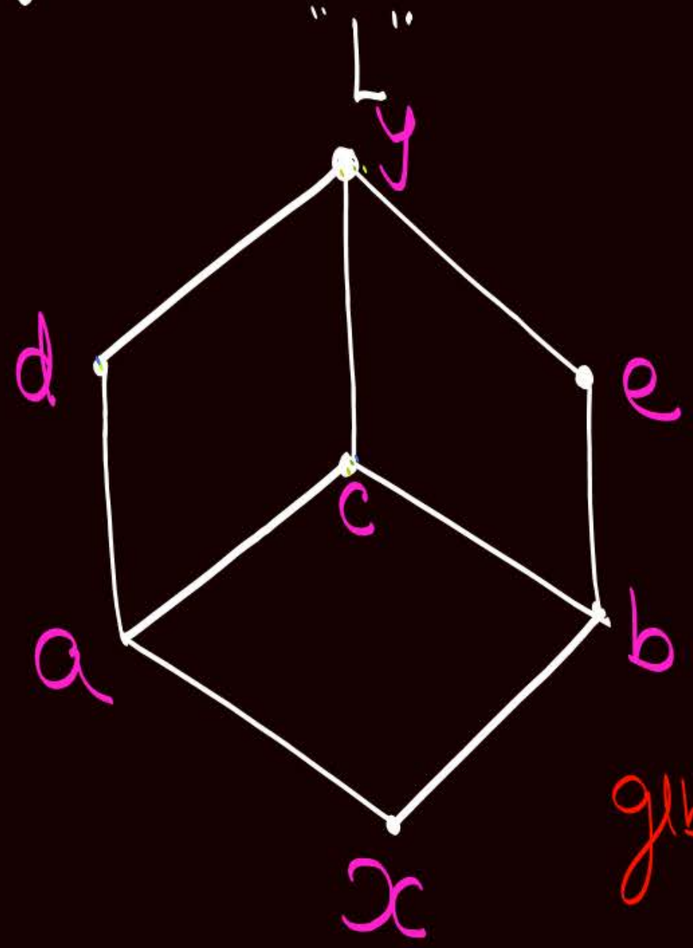
✓ ① It is a lattice

② But

lub(a,b) in L_1 = y ≠ lub(a,b) in L

∴ L_1 is not a sub-lattice of L

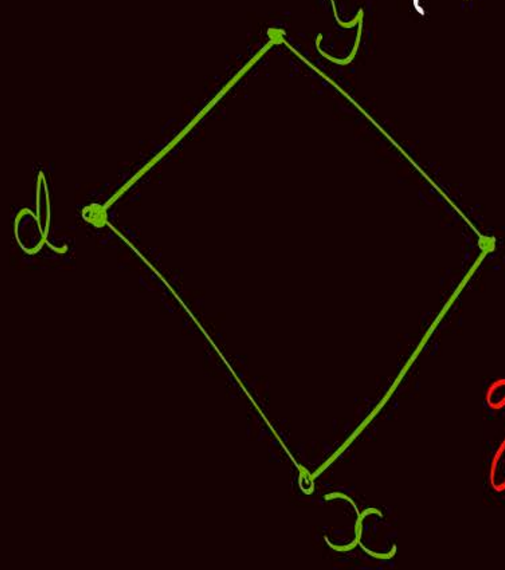
eg Consider the following Lattice 'L'



glb(c,d) = a

Which of the following is/are sublattices of lattice L

(ii) $L_2 = \{x, c, d, y\}$



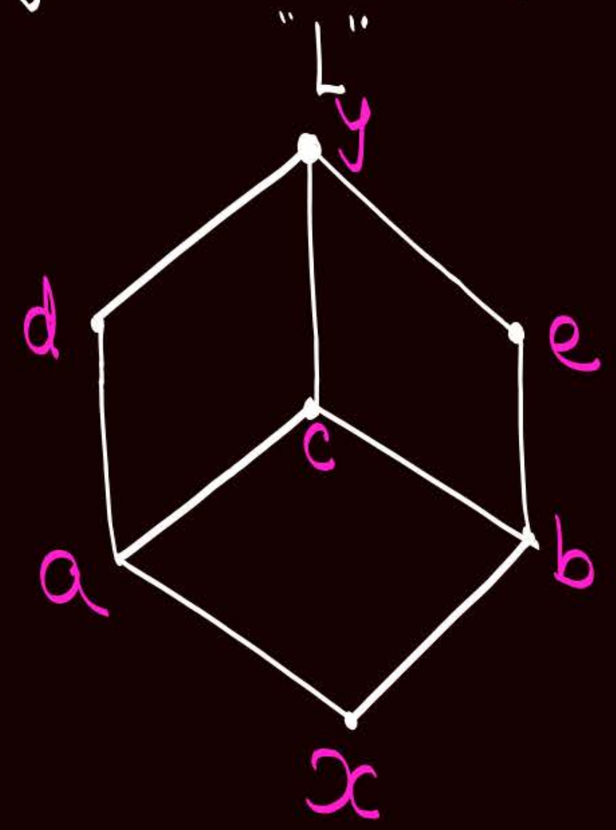
① It is a lattice

② But

glb(c,d) in L_2 = x ≠ glb(c,d) in L

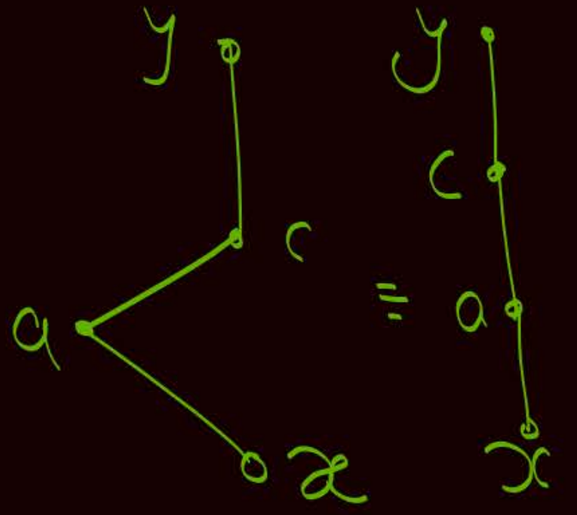
∴ Not a Sub-lattice of L

eg Consider the following lattice 'L'



Which of the following is/are sublattices of lattice L

(iii) $L_3 = \{x, a, c, y\}$

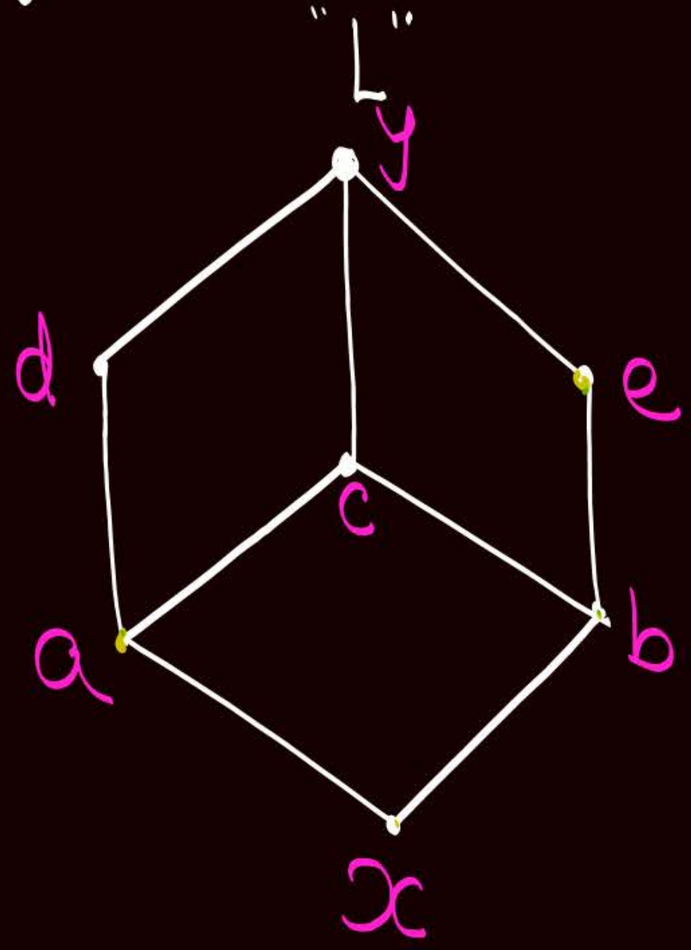


① It is a lattice

② lub & glb for every pair of elements in L_3 will be same as lub & glb for that pair of elements in L

∴ L_3 is a sub-lattice of L

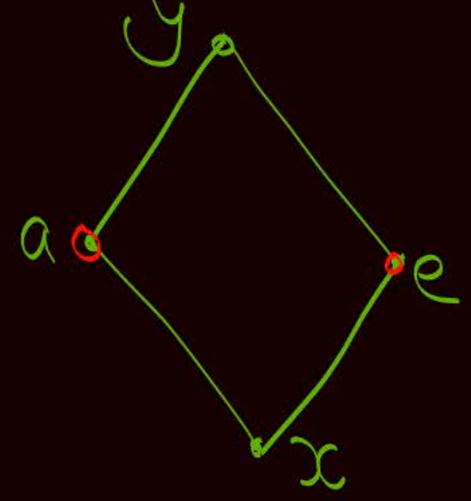
eg Consider the following lattice 'L'



lub(a,e) = y
glb(a,e) = x

Which of the following is/are sublattices of lattice L

(iv) $L_4 = \{x, a, e, y\}$



① It is a lattice

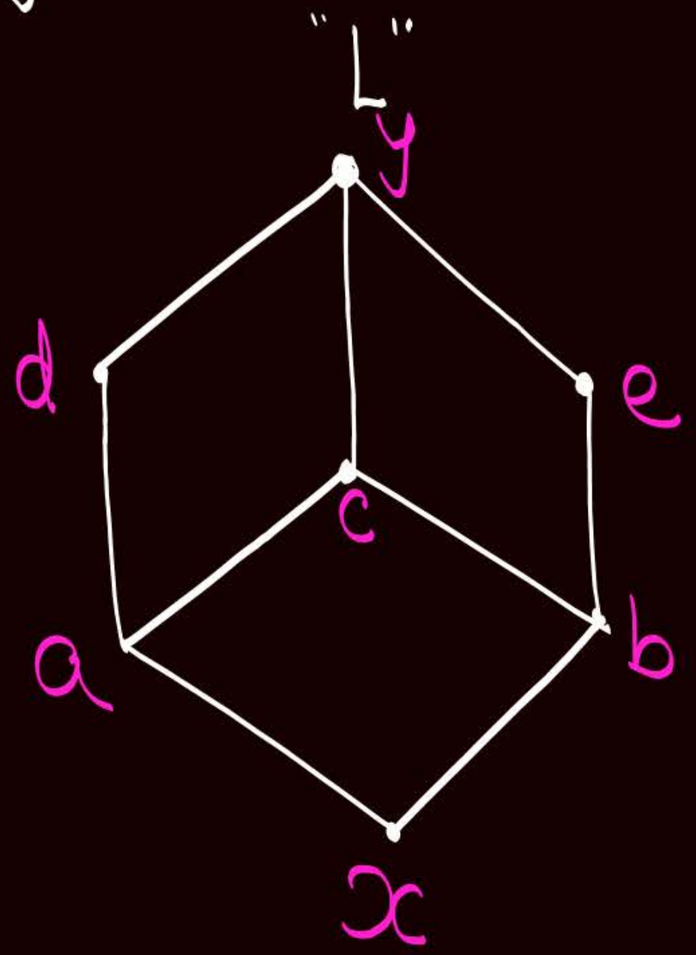
② $\underset{\text{in } L_4}{lub(a,e)} = y = lub(a,b) \text{ in } L$

&

$\underset{\text{in } L_4}{glb(a,e)} = x = glb(a,e) \text{ in } L$

$\therefore L_4$ is sub-lattice of L

eg Consider the following lattice 'L'



Which of the following is/are sublattices of lattice L

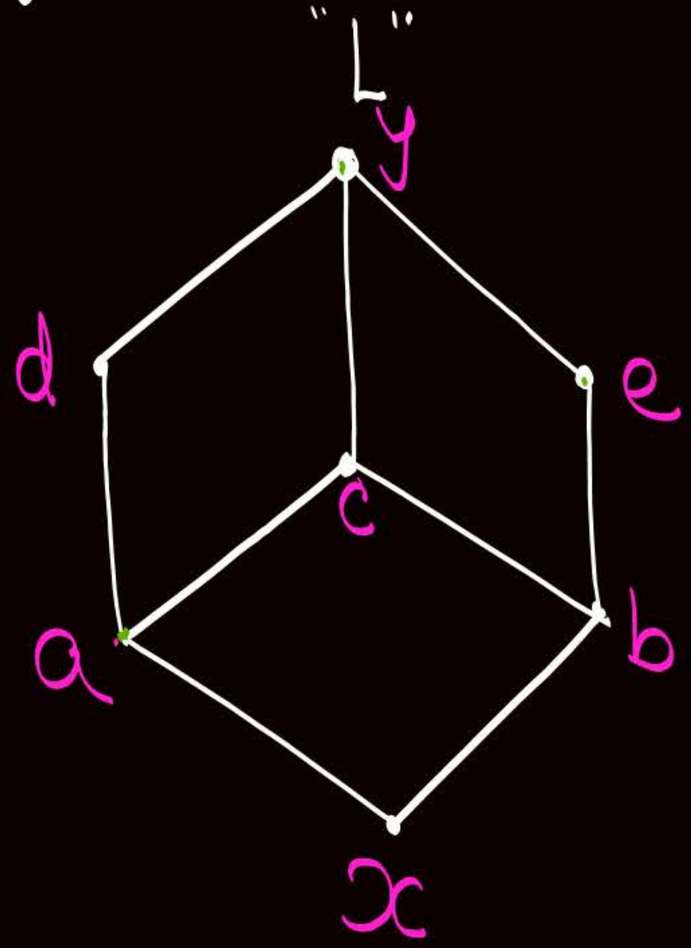
(i) $L_1 = \{x, a, b, y\}$

(ii) $L_2 = \{x, c, d, y\}$

(iii) $L_3 = \{x, a, c, y\}$
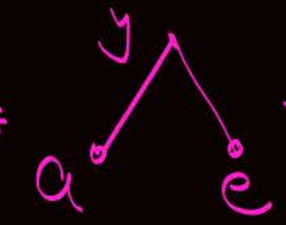
(iv) $L_4 = \{x, a, e, y\}$

eg Consider the following lattice 'L'



Which of the following is/are sub-lattices of lattice L

(a) $L_1 = \{a, e, y\}$:



← it is not a lattice

(b) $L_2 = \{a, b, c\}$:



← it is not a lattice

(c) $L_3 = \{x, a, d\}$



← it is a lattice ✓
← on the same line so lub & glb also same

(d) $L_4 = \{a, c, y\}$



← it is a lattice ✓
← on the same line so lub & glb also same

Today's Topic



Topic : Distributive lattice

* Let $[L, \vee, \wedge]$ be a lattice.

$[L, \vee, \wedge]$ is a distributive lattice if and only if,

(i) $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$

and (ii) $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ $\leftarrow \forall a, b, c \in L$

* Lattice is called a distributive lattice if and only if distributive property is satisfied by every combination of elements of the lattice

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Note:-

① In a distributive every element has at most one complement

If lattice is a distributive lattice, then every element will have at most one complement

but converse of the statement need not be true

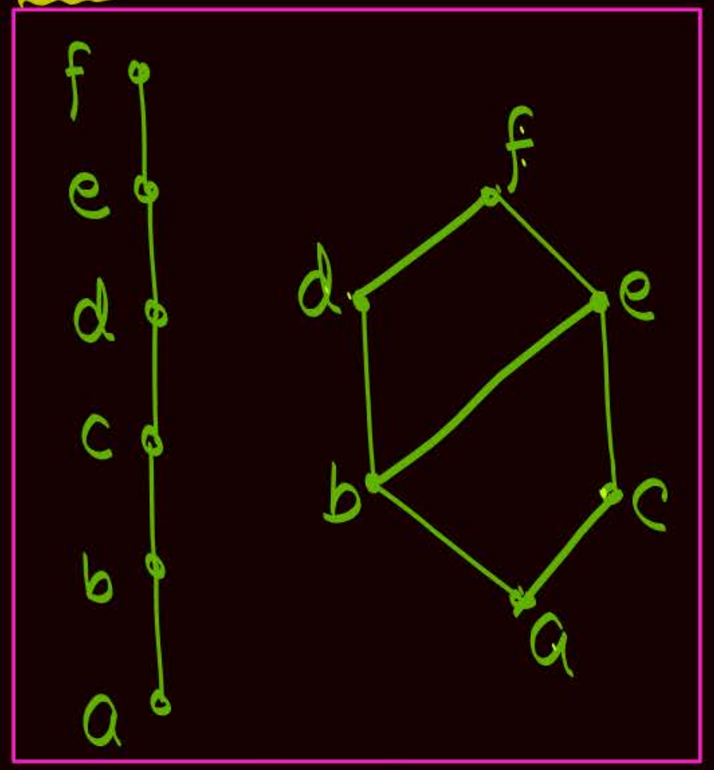
Note:-

② In a lattice, If any element of the lattice has more than one complement, then that lattice is not a distributive lattice

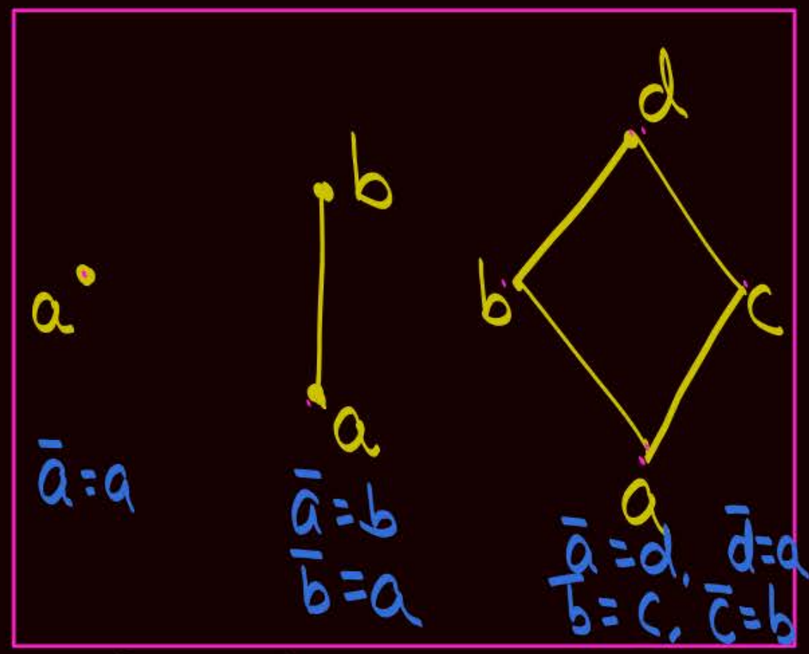
③ In a lattice, if Every element has exactly '1' Complement, then that lattice is definitely a distributive lattice

④ In a lattice, if Every element has at most one Complement, 'but not Exactly one'; then that lattice may or may not be distributive

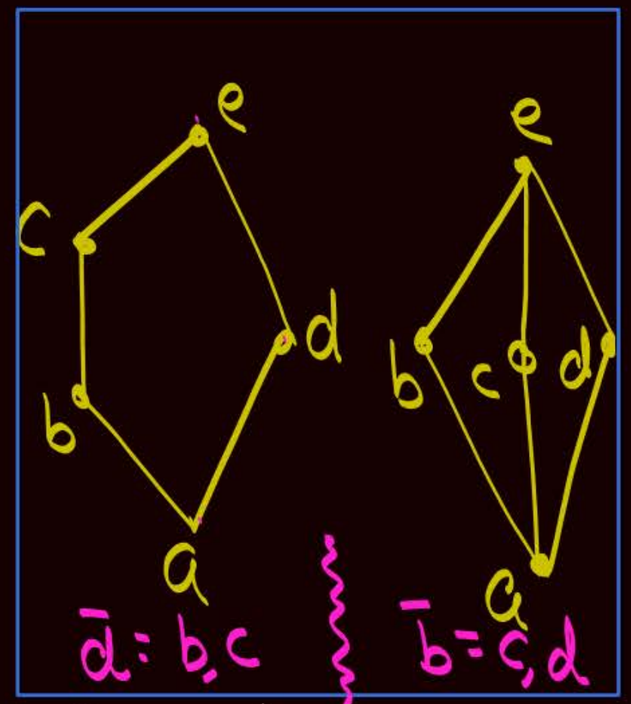
Note:-



* Examples in which Every element has '0' or '1' Complement
∴ We need to check for distributive property



* Every element has Exactly one Complement.
∴ Above lattice are distributive lattice



Some elements have more than On Complement
∴ Above lattice are not distributive lattice

Note :→

① If we choose elements from the same line of "Hasse diagram", then distributive property will always be satisfied by those elements.

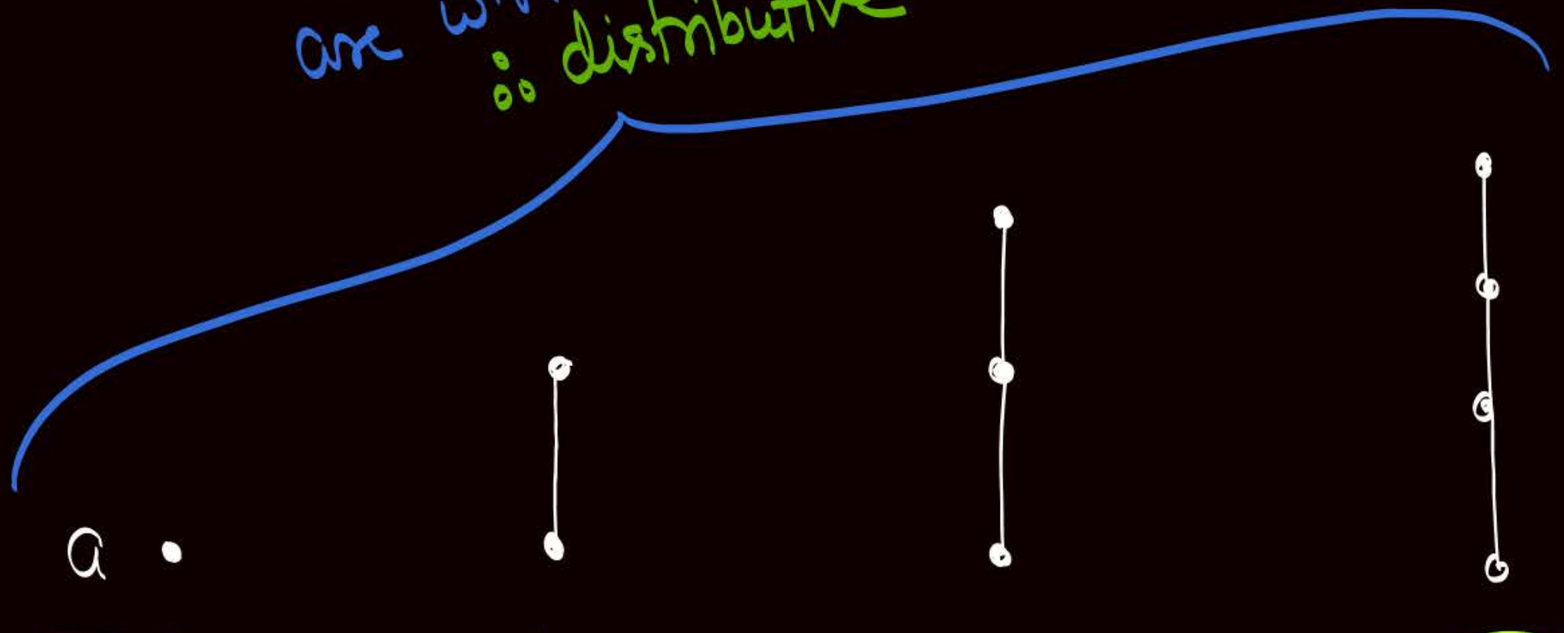


② If Lattice is a TOSET, then distributive Property will always be satisfied by the elements of that lattice

∴ A TOSET is always a distributive lattice

Note:

These '4' lattices are w.r.t. TOSET
∴ distributive



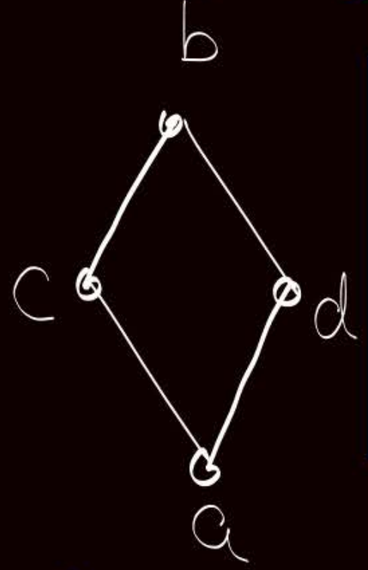
Hasse diagram w.r.t lattice of size = 1

Hasse diagram w.r.t lattice of size = 2

Hasse diagram w.r.t lattice of size = 3

Hasse diagrams w.r.t lattice of size = 4

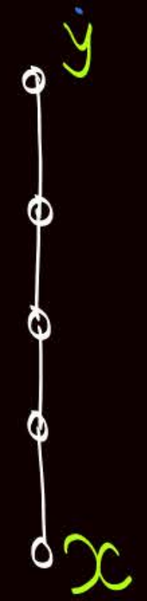
In this lattice Every element has Exactly '1' Complement ∴ distributive



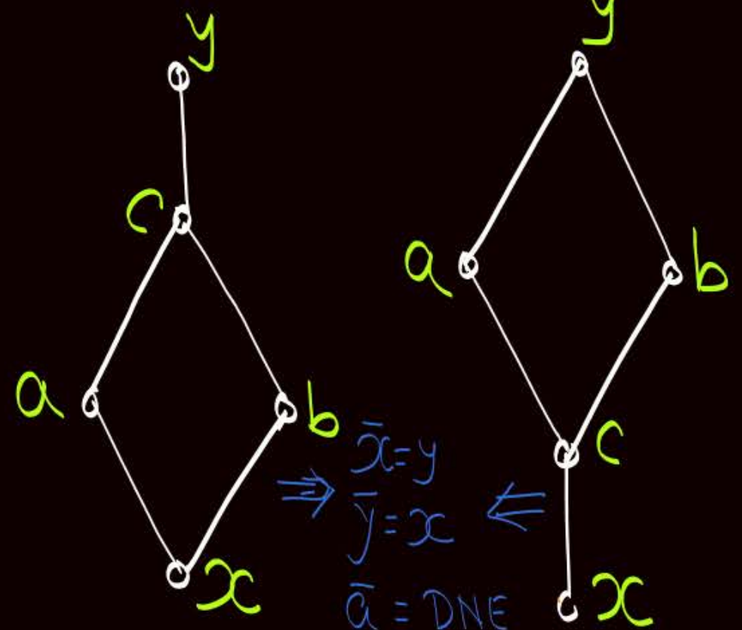
$\bar{a} = b$
$\bar{b} = a$
$\bar{c} = d$
$\bar{d} = c$

Note: Any lattice with '4' or less elements is always distributive

Note: Hasse diagrams possible w.r.t lattice of size = 5

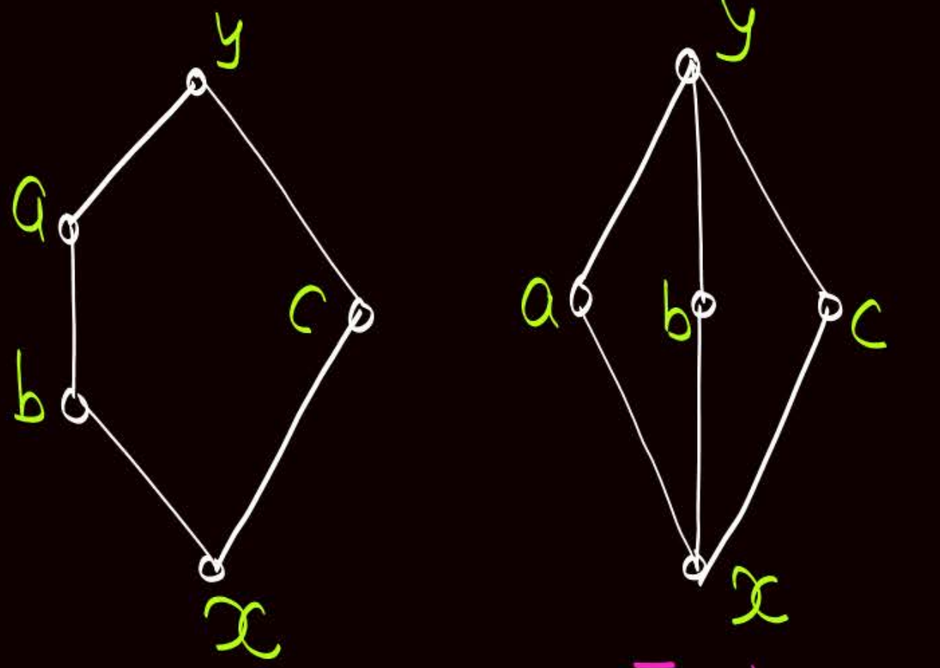


TOSET
∴ distributive



$\bar{x} = y$
$\bar{y} = x$
$\bar{a} = DNE$
$\bar{b} = DNE$
$\bar{c} = DNE$

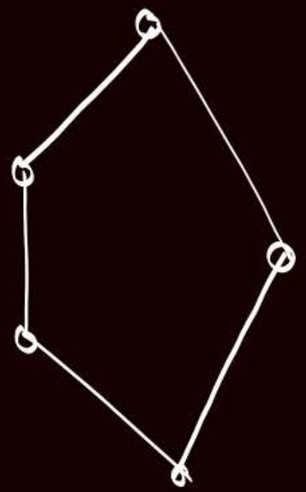
Every element has at most 1 complement, ∴ may or may not be distributive; But above lattices are distributive



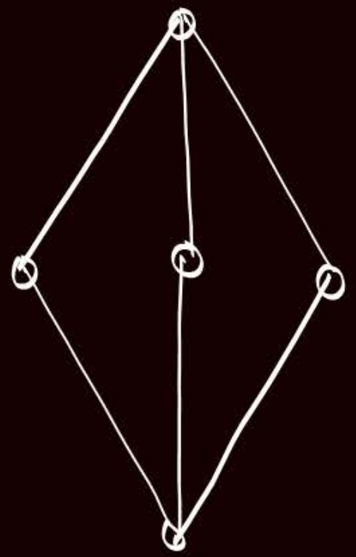
$\bar{c} = a, b$

$\bar{a} = b, c$

Some elements are having more than one complement
∴ Not distributive



L_1^*



L_2^*

* L_1^* and L_2^* are only two lattice of size = 5 that are not distributive

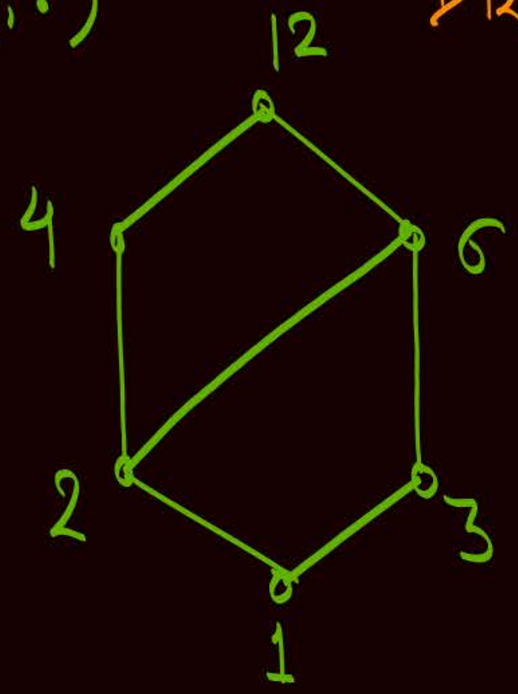
Note :→ A lattice L is not distributive if and only if there exist a sub-lattice of lattice L which is isomorphic to L_1^* or L_2^*

"imp"

(here the meaning is that looks similar)

Q:- Check whether the given lattice is distributive or not?

$(D_{12}, \div)$ $D_{12} = \{1, 2, 3, 4, 6, 12\}$



① No. of elements > 4, ∴ May or may not be distributive

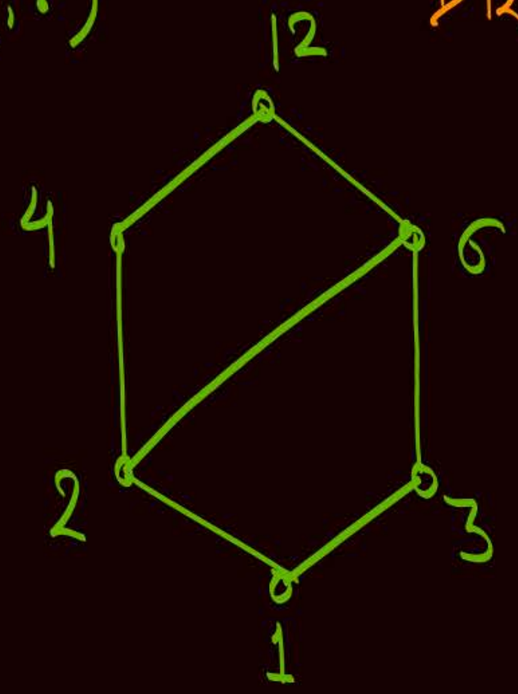
② Not a TOSET, ∴ May not be distributive

③ $\bar{1} = 6$
$\bar{12} = 1$
$\bar{2} =$ DNE
$\bar{3} = 4$
$\bar{4} = 3$
$\bar{6} =$ DNE

Every element has at most '1' Complement, ∴ May or may not be distributive.

Q:- Check whether the given lattice is distributive or not?

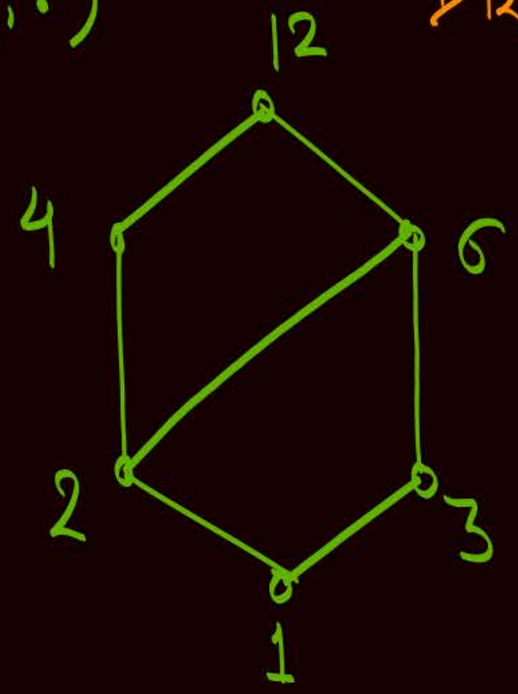
$(D_{12}, \div)$ $D_{12} : \{1, 2, 3, 4, 6, 12\}$



→ Do we have any subset of set $\{1, 2, 3, 4, 6, 12\}$, s.t the Hasse diagram is isomorphic to L_1^* or L_2^*

Q:- Check whether the given lattice is distributive or not?

$(D_{12}, \div)$ $D_{12} : \{1, 2, 3, 4, 6, 12\}$

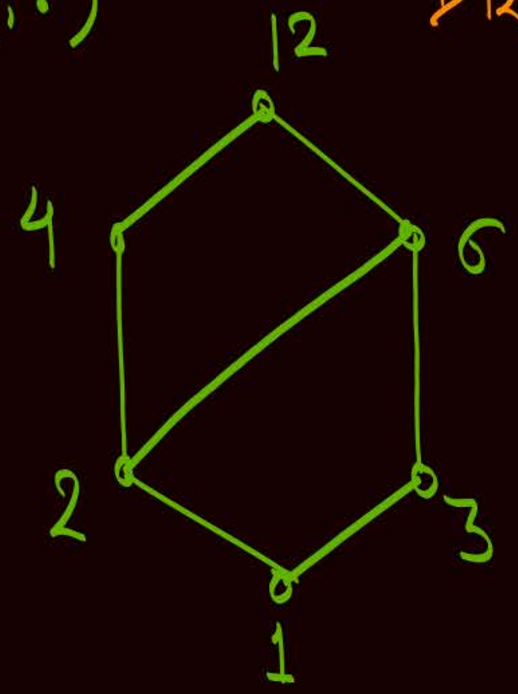


→ Do we have any subset of set $\{1, 2, 3, 4, 6, 12\}$, s.t the Hasse diagram is isomorphic to L_1^* or L_2^*

(1) → We can not choose any combination of '5' elements s.t Hasse diagram is isomorphic to L_2^*

Q:- Check whether the given lattice is distributive or not?

$(D_{12}, \div)$ $D_{12} : \{1, 2, 3, 4, 6, 12\}$

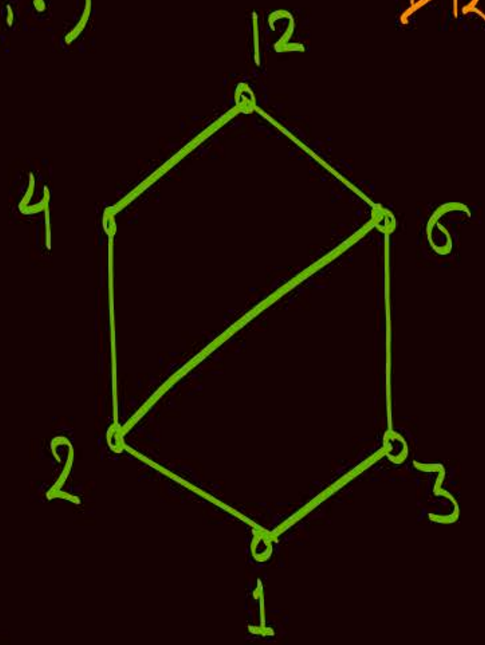


→ Do we have any subset of set $\{1, 2, 3, 4, 6, 12\}$, s.t the Hasse diagram is isomorphic to L_1^* or L_2^*

(2) $\{1, 2, 3, 4, 12\}$ & $\{1, 3, 4, 6, 12\}$ are two subsets s.t. Hasse diagram will be isomorphic to L_1^*

Q:- Check whether the given lattice is distributive or not?

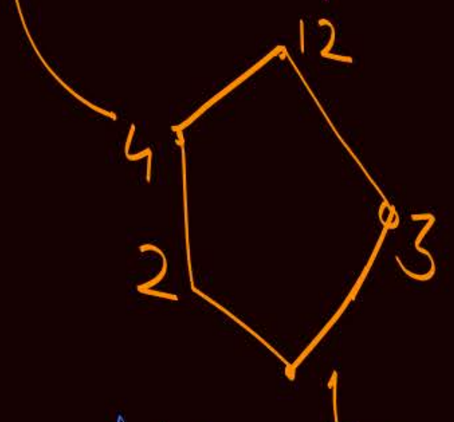
$(D_{12}, \div)$ $D_{12} = \{1, 2, 3, 4, 6, 12\}$



$lub(2,3) = 6$

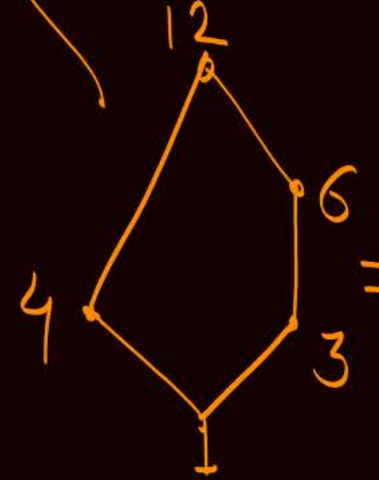
$glb(4,6) = 2$

② $\{1, 2, 3, 4, 12\}$ & $\{1, 3, 4, 6, 12\}$ are two subsets s.t. Hasse diagram will be isomorphic to L_1*

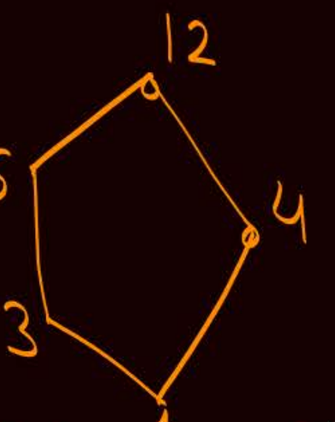


$lub(2,3) = 12 \neq$ original lattice

∴ it is not a sublattice of L



⇒

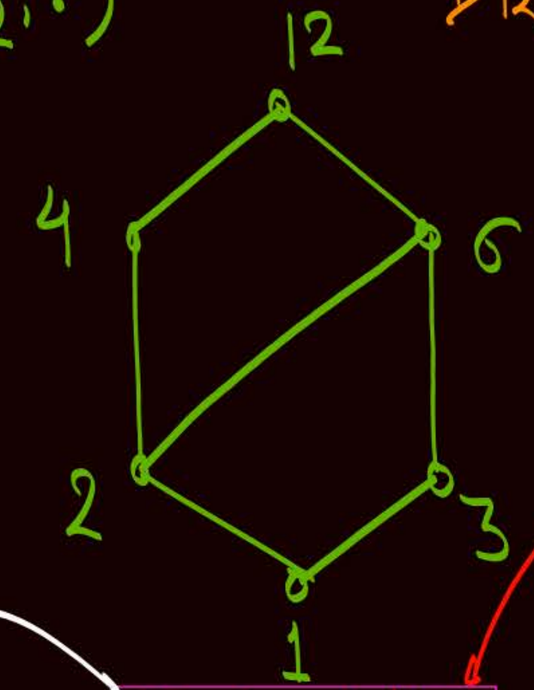


$glb(4,6) = 1 \neq$ original lattice

∴ it is not a sublattice of L

Q:- Check whether the given lattice is distributive or not?

$(D_{12}, \div)$ $D_{12} = \{1, 2, 3, 4, 6, 12\}$



→ Do we have any sub-lattice that is isomorphic to L_1^* or L_2^*

① No subset s.t. Hasse diagram is isomorphic to L_2^*, ∴ No sub-lattice is isomorphic to L_2^*

② We have two subsets $\{1, 2, 3, 4, 12\}$ & $\{1, 3, 4, 6, 12\}$ s.t. Hasse diagrams are isomorphic to L_1^* but none of them is a sub-lattice of given lattice

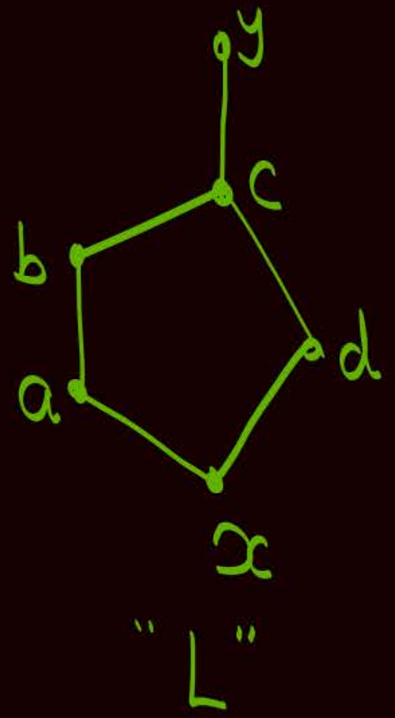
i.e. No sub-lattice which is isomorphic to L_1^*

imp

No sub-lattice which isomorphic to L_1^* or L_2^*

Hence, given lattice is a distributive lattice

Q:. Check whether the following lattice is a distributive lattice or not?



"L"

(i) No. of elements > 4, ∴ May or may not be distributive

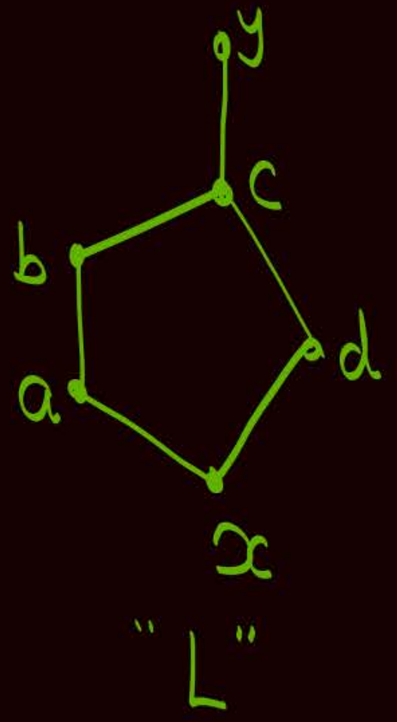
(ii) Not a TOSET, ∴ May or may not be distributive

(iii) $\bar{x} = y$
$\bar{y} = x$
$\bar{a} = DNE$
$\bar{b} = DNE$
$\bar{c} = DNE$
$\bar{d} = DNE$

Every Element has at most '1' Complement,

∴ May or may not be distributive

Q:. Check whether the following lattice is a distributive lattice or not?



"L"

$\bar{x} = y$
$\bar{y} = x$
$\bar{a} =$ DNE
$\bar{b} =$ DNE
$\bar{c} =$ DNE
$\bar{d} =$ DNE

← may or may not be distributive

But we have a sub-lattice $\{x, a, b, c, d\}$ which is isomorphic to L_1^*

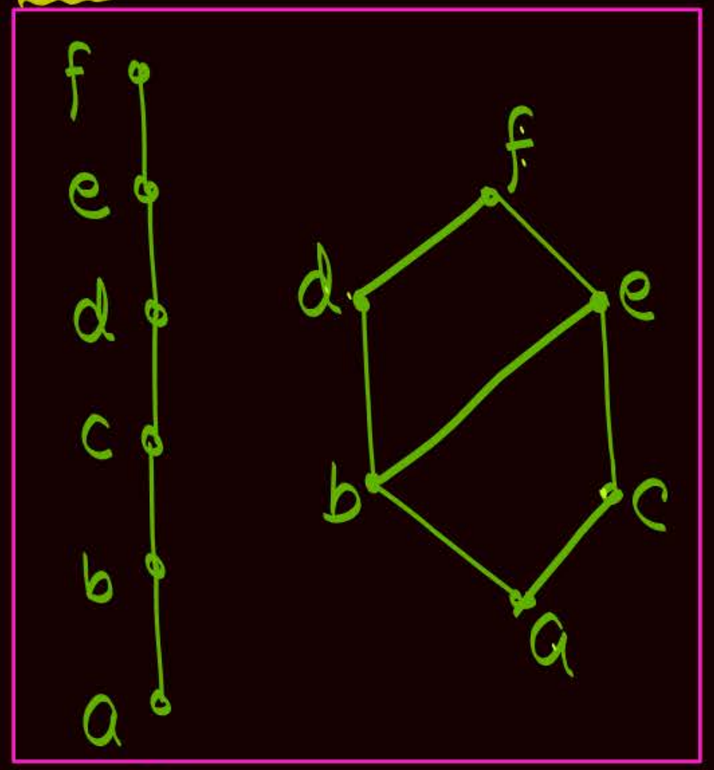
∴ Given lattice is not distributive

* Boolean Algebra / Boolean lattice

- A lattice in which every element has Exactly one complement is called a Boolean lattice

- A lattice which is Complemented as well as distributive is called a Boolean lattice

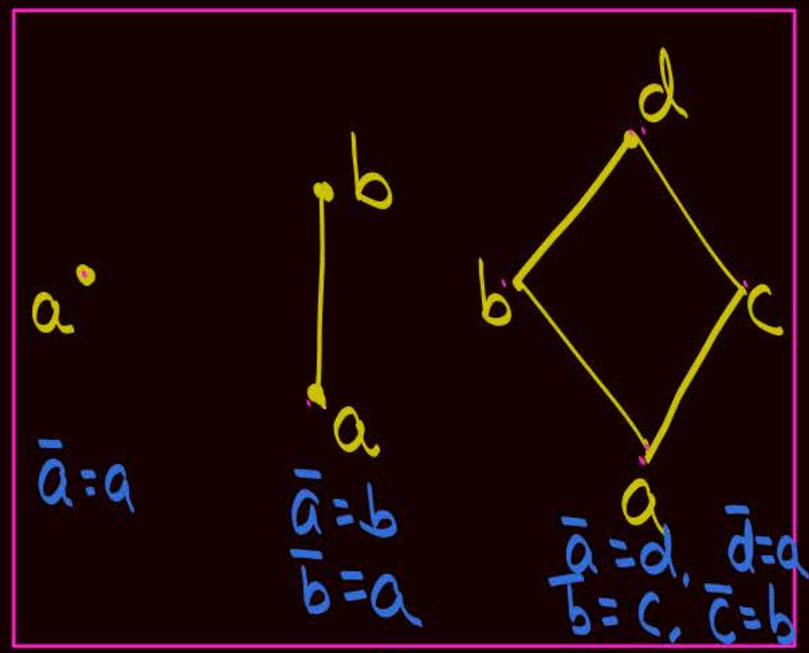
Note:-



Distributive, but not Complemented

⇓

∴ Not a Boolean lattice

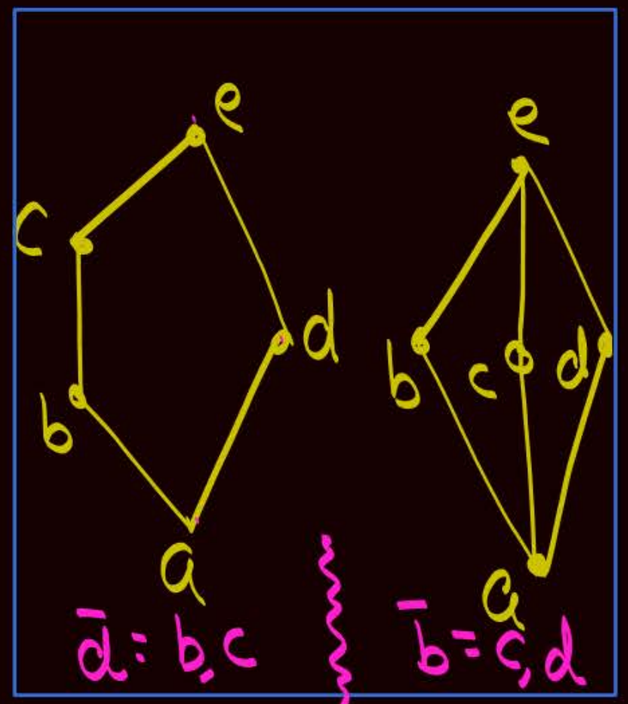


⇓

Complemented as well as distributive,

⇓

∴ Boolean lattice



⇓

Complemented lattice but not a distributive lattice

⇓

∴ Not a Boolean lattice

* **Note :→** If 'n' is positive integer such that D_n has no element which is a perfect square except '1', then 'n' is called a square free integer

D_n → Set of all Positive divisors of 'n'

eg: $D_{21} = \{1, 3, 7, 21\}$

No element (except 1) is a perfect square

∴ '21' is a square free integer

eg $D_{20} = \{1, 2, 4, 5, 10, 20\}$

'4' is a perfect square ∴ '20' is not a square free integer

Note: If n is a square free integer, then $(D_n, \div)$ is a Boolean lattice, and for an element $x \in D_n$, $\overline{x} = \frac{n}{x}$

* We know '21' is a square free integer

& $D_{21} = \{1, 3, 7, 21\}$.

then in Boolean lattice $\{D_{21}, \div\}$

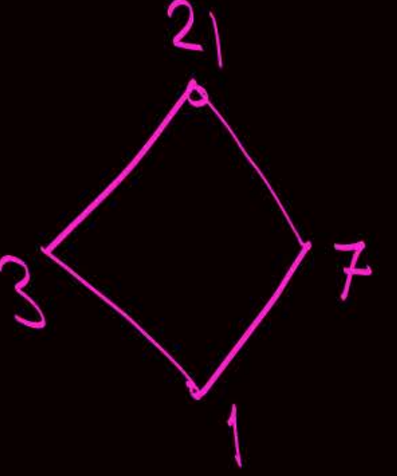
"21" is square free integer

$\therefore (D_{21}, \div)$ is a boolean lattice

& $\overline{1} = \frac{21}{1} = 21$, $\overline{3} = \frac{21}{3} = 7$

$\overline{7} = \frac{21}{7} = 3$, $\overline{21} = \frac{21}{21} = 1$

$\Longrightarrow$ Hasse diagram



$\overline{1} = 21$

$\overline{21} = 1$

$\overline{3} = 7$

$\overline{7} = 3$

Note:- ① If lattice is a Boolean Algebras then no. of Elements in lattice (Boolean Algebra) will always be in Power of '2': {i.e; $2^0, 2^1, 2^2, \ldots$} but Converse of the statement need not be true "and"

② If we have a Boolean algebra with 2^n elements, then number of Edges in Hasse diagram Corresponding to that Boolean Algebra will always be $n \cdot 2^{n-1}$

Note :→ Boolean lattice

Lattice of size $= 2^0 = 1$

a •

$\bar{a} = a$

Every element has exactly '1' Complement, ∴ Boolean lattice

$\#\text{Edges} = 0 \cdot 2^{0-1} = 0$

Lattice of size $= 2^1 = 2$

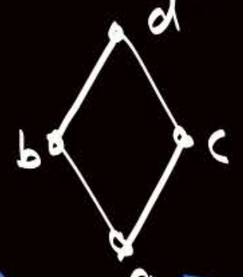


$\bar{a} = b$ & $\bar{b} = a$

Every element has exactly '1' Complement, ∴ Boolean lattice

$\#\text{Edges} = 1 \cdot 2^{1-1} = 1$

Lattice of size $= 2^2 = 4$



$\bar{a} = d$, $\bar{d} = a$

$\bar{b} = c$ & $\bar{c} = b$

Every element has exactly '1' Complement, ∴ Boolean lattice

$\#\text{Edges} = 2 \cdot 2^{2-1} = 4$



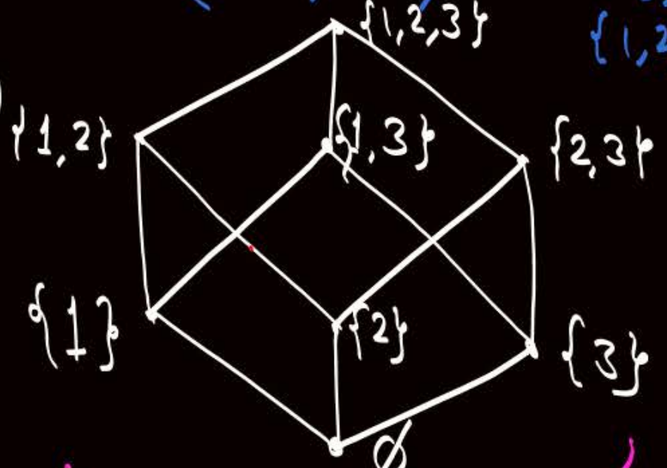
it is also a lattice of size: 4 but it is not a Boolean lattice

Lattice of size $= 2^3 = 8$

lattice $(P(A), \subseteq)$

let $A = \{1,2,3\}$

$P(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$



Let $X \in P(A)$ then $\bar{X} = A - X$

Unique Complement for Every element.

∴ Boolean lattice

$\#\text{Edges} = 3 \cdot 2^{3-1} = 12$

H.W.

Q.1 :

Which of the following statements is/ are not true

a) If A is any finite set then $[P(A), \subseteq]$ is distributive lattice
b) Every sub lattice of a distributive lattice is also a distributive lattice
c) Every totally ordered set is a distributive lattice
d) Every totally ordered set is bounded
e) Every distributive lattice is bounded
f) Every distributive lattice is a complemented lattice

Slide

H.W.

Which of the following lattice is /are not distributive?

(A) $(D_{125}, \div)$
(B) $(P(A), \subseteq)$
(C) $(D_{12}, \div)$
(D) $(\{1, 2,3,5,30\}, \div)$
(E) $(\{1, 2,3, 4,12\}, \div)$

Slide

H.W.

Which of the following lattice is /are not Complemented ?

(A) $(D_{125}, \div)$
(B) $(P(A), \subseteq)$
(C) $(D_{12}, \div)$
(D) $(\{1, 2,3,5,30\}, \div)$
(E) $(\{1, 2,3, 4,12\}, \div)$

Slide

H.W.

Which of the following lattice is /are boolean lattice ?

(A) $(D_{125}, \div)$
(B) $(P(A), \subseteq)$
(C) $(D_{12}, \div)$
(D) $(\{1, 2,3,5,30\}, \div)$
(E) $(\{1, 2,3, 4,12\}, \div)$

Slide



2 mins Summary

Topic Distributive lattice

Topic Boolean algebra

Slide

THANK - YOU